Bachelor's thesis

Machine Learning & High Frequency Financial Time Series

Predicting short-term regime shifts via Hidden Markov Models

Egil Furnes
egilsf@uio.no

Department of Mathematics
Faculty of Mathematics and Natural Sciences

28th March 2026

Abstract

High-frequency foreign exchange (FX) markets exhibit complex behaviors that challenge standard linear time series models. While daily returns often appear random, tick-by-tick data reveals significant short-term dependence driven by market microstructure effects. This project investigates the modeling of time-varying short-term dependence in FX data using a Regime-Switching Autoregressive model formulated as a Hidden Markov Model (HMM). By decomposing the observed price into a latent efficient price and microstructure noise, we apply pre-averaging techniques to mitigate noise before estimating dependence regimes. We aim to capture the switching behavior between noise-dominated periods and momentum-driven price runs, comparing the HMM approach against static baseline models and online adaptive estimators.

Acknowledgements

I would like to thank my supervisor, Associate Professor Gudmund Horn Hermansen (Department of Mathematics, University of Oslo), for his guidance and support in the fields of Statistics and Data Science throughout this project.

1 Introduction

Financial time series are well known to exhibit non-stationarity, volatility clustering, and structural change. While standard statistical learning techniques, such as those described by Hastie et al. (2009), provide robust frameworks for general data analysis, high-frequency financial data requires specialized treatment. As noted by Hautsch (2012), the analysis of tick-by-tick data introduces unique challenges related to the discreteness of price changes and the irregular spacing of transactions.

The presence of time-varying dependence in financial returns has been a subject of extensive research. Hamilton (1989) introduced a seminal framework for modeling regime changes in economic time series, and further expanded it in 1994 (Hamilton 1994). However, in the context of high-frequency market structure, Lo and MacKinlay (1990) and Engle (1982) identified that short-horizon return autocorrelation is not constant. Modern approaches often require robust estimation techniques; Jacod et al. (2017) and Aït-Sahalia and Jacod (2014) provide critical methodologies for separating microstructure noise from the efficient price. To model the latent states driving these market dynamics, we rely on the Hidden Markov Model (HMM) framework detailed by Rabiner (1989), while also considering the broader financial time series context provided by Tsay (2010). HMM-based approaches have proven effective in intraday momentum trading, where latent states capture shifts between trending and mean-reverting regimes (Christensen et al. 2020), and in FX markets specifically, where coupling hidden states across correlated instruments has been shown to improve trading signals (Lee 2013).

1.1 Bid–ask bounce

The bid-ask bounce is a fundamental microstructure phenomenon where transaction prices oscillate between the bid and ask quotes even if the underlying asset value remains unchanged. This bouncing effect induces a negative autocorrelation in returns

at the highest frequencies, creating a "noise" regime that can obscure genuine price trends. Zabaljauregui and Campi (2020) formalise this intuition within a stochastic control framework, showing that a market maker facing unobservable regime shifts must widen spreads to compensate for the additional regime uncertainty—a finding that directly motivates modelling the latent microstructure state rather than conditioning on observable proxies alone.

1.2 Order flow imbalance

Order flow imbalance refers to the disparity between aggressive buy and sell orders. When order flow is heavily skewed in one direction, it can deplete liquidity at the best quote, causing the price to move. Unlike the mean-reverting bid-ask bounce, strong order flow imbalance can induce short-term momentum or positive serial dependence in returns.

1.3 Asynchronous trading

Asynchronous trading occurs because assets trade at different frequencies and times. Lo and MacKinlay (1990) demonstrated that this non-synchronicity can induce spurious cross-autocorrelations between assets. In the context of a single FX pair, irregular arrival times of information and trades contribute to the stochastic nature of the volatility and dependence structure we aim to model.

Paper outline

The remainder of the paper proceeds as follows. Section 2 develops the modelling framework: we begin with the price decomposition and pre-averaging procedure, define the baseline AR(1) benchmark, and then specify the full Hidden Markov Model with regime-switching autoregressive dynamics, including volatility extensions. We derive the key inference algorithms—the Forward algorithm for likelihood evaluation, the Viterbi algorithm for optimal state decoding, and the Baum–Welch expectation–maximisation procedure for parameter estimation—and describe the model selection criterion used to choose the number of regimes. Section 3 presents exploratory data analysis of the EUR/USD tick data and preliminary empirical findings. Section 4 summarises the project timeline and outlines remaining work.

2 Method

Let P_τ denote the observed transaction price at tick time τ. The observed price can be decomposed as shown in Equation 1:

$$P_\tau = P_\tau^* + \eta_\tau \tag{1}$$

where P_τ^* is the latent efficient price and η_τ represents microstructure noise. The pre-averaged price is defined in Equation 2 as

$$\bar{P}_t = \frac{1}{k}\sum_{j=0}^{k-1} P_{t-j} \tag{2}$$

where t now indexes the last tick in each averaging block. Returns are then constructed as shown in Equation 3:

$$r_t = \bar{P}_t - \bar{P}_{t-1} \tag{3}$$

2.1 Baseline Auto-regressive Model

As a benchmark, we consider a single-regime auto-regressive model of order one, given by Equation 4:

$$r_t = \mu + \phi r_{t-1} + \varepsilon_t \tag{4}$$

where μ is the unconditional mean return, ϕ measures linear serial dependence and ε_t is an innovative term with $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$.

2.2 Hidden Markov Model with Regime-Switching Auto-regressive Dynamics

Let $S_t \in \{1, \ldots, K\}$ denote an unobserved discrete-time Markov chain representing the dependence regime at time t. The transition probabilities are defined in Equation 5:

$$\mathbb{P}(S_t = j \mid S_{t-1} = i) = p_{ij} \tag{5}$$

with $p_{ij} \geq 0$ and $\sum_{j=1}^{K} p_{ij} = 1$ for all i.

Conditional on the regime $S_t = k$, returns follow an auto-regressive process described by Equation 6:

$$r_t = \mu_k + \phi_k r_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_k^2) \tag{6}$$

Here μ_k is the regime-specific mean, ϕ_k captures the strength of the short-term dependence in regime k and σ_k^2 is the regime-specific innovation variance.

Regimes with ϕ_k close to zero are interpreted as noise-dominated periods, while regimes with larger positive ϕ_k correspond to short-term momentum or price returns.

2.3 Modeling volatility

Allowing σ_k^2 to vary across regimes captures abrupt changes in market uncertainty. As an extension, conditional heteroskedasticity can be introduced via a regime-dependent ARCH specification, shown in Equation 7:

$$\sigma_{k,t}^2 = \omega_k + \alpha_k \varepsilon_{t-1}^2 \tag{7}$$

where $\omega_k \geq 0$ and $\alpha_k \geq 0$. Building on this, volatility persistence can be incorporated through a regime-dependent GARCH specification, see Equation 8:

$$\sigma_{k,t}^2 = \omega_k + \alpha_k \varepsilon_{t-1}^2 + \beta_k \sigma_{k,t-1}^2, \tag{8}$$

where $\beta_k \geq 0$. This extension allows volatility shocks to decay gradually over time within each regime rather than affecting only the next period.

2.4 Complete parameter set and emission density

The complete set of HMM parameters is

$$\theta = \{\pi,\, \mathbf{P},\, \mu_k,\, \phi_k,\, \sigma_k^2 : k = 1, \ldots, K\}, \tag{9}$$

where $\pi = (\pi_1, \ldots, \pi_K)^\top$ with $\pi_k = \mathbb{P}(S_1 = k)$ is the initial state distribution, $\mathbf{P} = (p_{ij})$ is the $K \times K$ transition matrix, and μ_k, ϕ_k, σ_k^2 are the regime-specific AR parameters introduced in Equation 6. Conditional on $S_t = k$ and the previous return r_{t-1}, the emission density is

$$f_k(r_t \mid r_{t-1}; \theta) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(r_t - \mu_k - \phi_k r_{t-1})^2}{2\sigma_k^2} \right). \tag{10}$$

2.5 Likelihood and the forward algorithm

Direct maximisation of the observed-data log-likelihood $\ell(\theta) = \log \mathbb{P}(r_{1:T}; \theta)$ is intractable because it requires summing over all K^T possible state sequences. The *forward algorithm* (Rabiner 1989; Cappé et al. 2005) computes the likelihood recursively. Define the forward variable

$$\alpha_t(k) = \mathbb{P}(r_1, \ldots, r_t,\, S_t = k \mid \theta), \tag{11}$$

initialised as $\alpha_1(k) = \pi_k f_k(r_1)$ and propagated via

$$\alpha_t(k) = \left[\sum_{i=1}^{K} \alpha_{t-1}(i)\, p_{ik}\right] f_k(r_t \mid r_{t-1}; \theta), \quad t = 2, \ldots, T. \tag{12}$$

The observed-data likelihood then equals $\mathbb{P}(r_{1:T}; \theta) = \sum_{k=1}^{K} \alpha_T(k)$.

2.6 Viterbi decoding

Given estimated parameters $\hat{\theta}$, the most probable latent state sequence is

$$\hat{S}_{1:T} = \arg\max_{S_{1:T}} \mathbb{P}(S_{1:T} \mid r_{1:T}; \hat{\theta}). \tag{13}$$

The *Viterbi algorithm* (Rabiner 1989) solves this by dynamic programming. Define

$$\delta_t(k) = \max_{S_{1:t-1}} \mathbb{P}(S_{1:t-1},\, S_t = k,\, r_{1:t} \mid \hat{\theta}), \tag{14}$$

initialised as $\delta_1(k) = \pi_k f_k(r_1)$ and updated via

$$\delta_t(k) = \max_{1 \le i \le K} \left[\delta_{t-1}(i)\, p_{ik}\right] f_k(r_t \mid r_{t-1}; \hat{\theta}). \tag{15}$$

Storing the maximising state at each step, the optimal path $\hat{S}_{1:T}$ is recovered by backtracking from $\hat{S}_T = \arg\max_k \delta_T(k)$.

2.7 Parameter estimation via Baum–Welch

Parameters are estimated by maximum likelihood using the *Baum–Welch algorithm* (Cappé et al. 2005), an instance of the Expectation–Maximisation (EM) procedure of Dempster et al. (1977). Alongside the forward variable $\alpha_t(k)$, define the backward variable

$$\beta_t(k) = \mathbb{P}(r_{t+1}, \ldots, r_T \mid S_t = k,\, \theta), \tag{16}$$

initialised as $\beta_T(k) = 1$ and propagated as $\beta_t(k) = \sum_{j=1}^{K} p_{kj}\, f_j(r_{t+1} \mid r_t; \theta)\, \beta_{t+1}(j)$. The E-step computes the smoothed state occupancy probabilities

$$\gamma_t(k) = \mathbb{P}(S_t = k \mid r_{1:T}; \theta) = \frac{\alpha_t(k)\, \beta_t(k)}{\displaystyle\sum_{j=1}^{K} \alpha_t(j)\, \beta_t(j)} \tag{17}$$

and the pairwise transition probabilities

$$\xi_t(i, j) = \mathbb{P}(S_{t-1} = i,\, S_t = j \mid r_{1:T}; \theta) \;\propto\; \alpha_{t-1}(i)\, p_{ij}\, f_j(r_t \mid r_{t-1}; \theta)\, \beta_t(j). \tag{18}$$

The M-step updates the parameters in closed form. Transition probabilities are updated as

$$p_{ij}^{\text{new}} = \frac{\sum_{t=2}^{T} \xi_t(i,j)}{\sum_{t=2}^{T} \gamma_{t-1}(i)}, \tag{19}$$

and the regime-specific AR parameters $(\mu_k, \phi_k, \sigma_k^2)$ are updated via weighted least squares with $\gamma_t(k)$ as observation weights. The E- and M-steps are iterated to convergence of the log-likelihood.

2.8 Model selection

The number of regimes K is treated as a hyperparameter. For each candidate K we compute the Bayesian Information Criterion

$$\text{BIC}(K) = -2\,\ell(\hat{\theta}_K) + p_K \log T, \tag{20}$$

where $p_K = K^2 + 3K - 1$ counts the free parameters (transition matrix entries, regime-specific AR intercepts, slopes, and variances, and initial distribution) and T is the number of observations. The preferred model minimises $\text{BIC}(K)$. The stronger complexity penalty of BIC relative to AIC is appropriate given the large sample sizes typical of tick data. Empirical evidence from HMM-based intraday trading strategies suggests that a small number of regimes, typically $K \in \{2, 3\}$, is sufficient to capture the dominant market dynamics (Christensen et al. 2020), providing prior support for evaluating candidate models in this range.

2.9 Data

Data was retrieved from HistData.com (2026).

2.10 Financial metrics

To evaluate the performance of the trading strategy, a set of financial metrics is computed. These metrics capture execution quality, trade-level performance, and portfolio-level risk and return characteristics.

2.10.1 Execution metrics

Average slippage measures the average absolute difference between the executed price and the intended price.

$$S_{\text{avg}} = \frac{1}{N} \sum_{i=1}^{N} |P_{\text{exec},i} - P_{\text{int},i}| \tag{21}$$

Fill rate measures the fraction of the requested trading volume that was actually executed.

$$\text{Fill Rate} = \frac{\sum_{i=1}^{N} V_{\text{exec},i}}{\sum_{i=1}^{N} V_{\text{req},i}} \tag{22}$$

2.10.2 Trade-level metrics

The average holding time measures the average time a position remains open.

$$\overline{T} = \frac{1}{N} \sum_{i=1}^{N} (t_{\text{close},i} - t_{\text{open},i}) \tag{23}$$

The win rate measures the fraction of trades that are profitable.

$$W = \frac{N_{\text{win}}}{N_{\text{total}}} \tag{24}$$

Profit factor measures the ratio of total profits to total losses.

$$\text{PF} = \frac{\sum \text{Gross Profits}}{\sum |\text{Gross Losses}|} \tag{25}$$

Average net profit per trade after transaction costs is defined as

$$\text{PnL}_{\text{avg}} = \frac{1}{N} \sum_{i=1}^{N} ((P_{\text{exit},i} - P_{\text{entry},i})Q_i - c_i) \tag{26}$$

Cumulative net profit and loss represents the total economic value of the strategy rather than just per-trade averages.

$$\text{PnL}_{\text{tot}} = \sum_{i=1}^{N} ((P_{\text{exit},i} - P_{\text{entry},i})Q_i - c_i) \tag{27}$$

Average trade return as a fraction of capital committed is given by

$$\bar{r}_{\text{trade}} = \frac{1}{N} \sum_{i=1}^{N} \frac{(P_{\text{exit},i} - P_{\text{entry},i})Q_i - c_i}{P_{\text{entry},i}Q_i} \tag{28}$$

The average gain and average loss separate the magnitude of winning and losing trades.

$$\overline{\text{Gain}} = \frac{1}{N_{\text{win}}} \sum_{i \in \mathcal{W}} \text{PnL}_i \tag{29}$$

$$\overline{\text{Loss}} = \frac{1}{N_{\text{loss}}} \sum_{i \in \mathcal{L}} |\text{PnL}_i| \tag{30}$$

Expectancy combines win rate and win/loss magnitude to measure expected profit per trade.

$$\text{Expectancy} = W \cdot \overline{\text{Gain}} - (1 - W) \cdot \overline{\text{Loss}} \tag{31}$$

2.10.3 Risk and return metrics

The maximum drawdown measures the largest peak-to-trough decline in portfolio value over the period.

$$\text{MDD} = \max_{\tau \in (0,T)} \left(\max_{t \in (0,\tau)} \frac{V(t) - V(\tau)}{V(t)} \right) \tag{32}$$

The Sharpe ratio measures excess return per unit of total volatility.

$$\text{Sharpe} = \frac{\mathbb{E}[R_p - R_f]}{\sigma_p} \tag{33}$$

The Sortino ratio measures the average excess return per unit of downside volatility.

$$\text{Sortino} = \frac{\mathbb{E}[R_p - R_f]}{\sigma_d} \tag{34}$$

Return volatility reports the variability of returns directly.

$$\sigma_R = \sqrt{\frac{1}{T-1} \sum_{t=1}^{T} (R_t - \bar{R})^2} \tag{35}$$

where

$$\bar{R} = \frac{1}{T} \sum_{t=1}^{T} R_t \tag{36}$$

The Calmar ratio relates returns to drawdown risk.

$$\text{Calmar} = \frac{\mathbb{E}[R_p - R_f]}{\text{MDD}} \tag{37}$$

Alternatively, it may be defined using annualized return:

$$\text{Calmar} = \frac{R_{\text{ann}}}{\text{MDD}} \tag{38}$$

2.10.4 Other metrics

Turnover measures how intensively the strategy trades relative to available capital, where A_0 is the initial capital.

$$\text{Turnover} = \frac{\sum_{i=1}^{N} |Q_i| P_{\text{entry},i}}{A_0} \tag{39}$$

Exposure time measures the fraction of total sample time during which the strategy has an open position.

$$\text{Exposure} = \frac{\sum_{i=1}^{N} (t_{\text{close},i} - t_{\text{open},i})}{T_{\text{sample}}} \tag{40}$$

3 Result

Figure 1 presents an exploratory data analysis of EUR/USD tick data for January 2026. The top panel shows that the mid-price evolves broadly continuously but with intermittent jumps, while the bid-ask spread is highly concentrated at the minimum tick size with an extended right tail characteristic of illiquid periods. The middle panel reveals strong intraday seasonality in the spread, with compression during the overlapping London–New York session and widening during the Asian session and around macroeconomic announcements; the distribution of one-minute log returns exhibits the heavy tails and excess kurtosis typical of high-frequency data. The bottom panel confirms that trading intensity peaks at market open and at the US open, and that annualised daily realised volatility fluctuates substantially across the month—evidence of the time-varying dependence structure that the HMM framework is designed to capture.

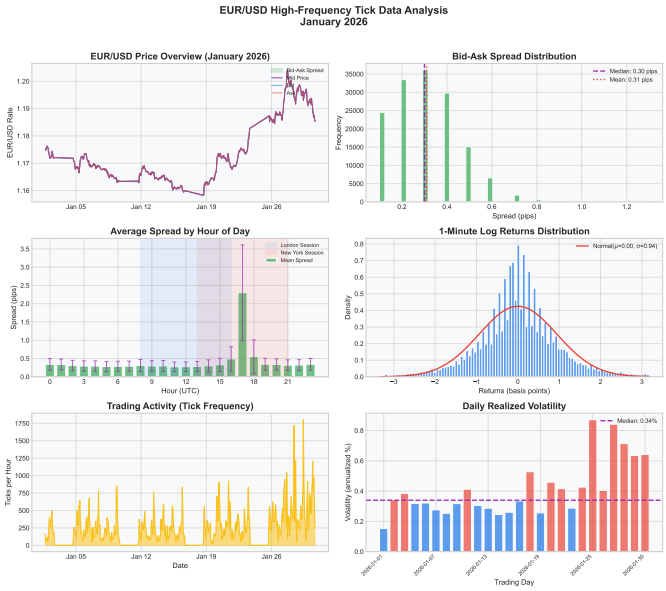

Figure 1: Exploratory data analysis of EUR/USD tick data for January 2026. The panels display the price evolution and bid-ask spread distribution (top); intraday spread seasonality highlighting the London and New York sessions, alongside the leptokurtic distribution of 1-minute log returns compared to a Gaussian benchmark (middle); and hourly trading activity frequency alongside annualized daily realized volatility (bottom).

4 Project Plan

- **February:** Literature review on HMMs and high-frequency econometrics. Data acquisition from HistData (EUR/USD) and implementation of pre-averaging techniques to handle microstructure noise.
- **March:** Implementation of the baseline single-regime Autoregressive (AR) model. Development of the Hidden Markov Model (HMM) with regime-switching AR dynamics in Python/R.
- **April:** Extension of the model to include regime-dependent volatility (ARCH/GARCH effects). Calibration and validation of the model against the baseline using in-sample and out-of-sample testing.

- **May (1–15):** Final interpretation of regime-switching results (identifying noise vs. momentum states), final write-up of the thesis, and preparation of the code repository.

References

Aït-Sahalia, Yacine and Jean Jacod (2014). *High-Frequency Financial Econometrics*. Princeton University Press. URL: https://press.princeton.edu/books/hardcover/9780691161433/high-frequency-financial-econometrics.

Cappé, Olivier, Eric Moulines and Tobias Rydén (2005). *Inference in Hidden Markov Models*. Springer. DOI: 10.1007/b136369. URL: https://doi.org/10.1007/b136369.

Christensen, Hugh, Simon Godsill and Richard E. Turner (2020). *Hidden Markov Models Applied to Intraday Momentum Trading with Side Information*. arXiv: 2006.08307 [q-fin.TR]. URL: https://arxiv.org/abs/2006.08307.

Dempster, Arthur P., Nan M. Laird and Donald B. Rubin (1977). 'Maximum Likelihood from Incomplete Data via the EM Algorithm'. In: *Journal of the Royal Statistical Society: Series B* 39.1, pp. 1–38. DOI: 10.1111/j.2517-6161.1977.tb01600.x. URL: https://doi.org/10.1111/j.2517-6161.1977.tb01600.x.

Engle, Robert F. (1982). 'Autoregressive Conditional Heteroskedasticity with Estimates of the Variance of United Kingdom Inflation'. In: *Econometrica* 50.4, pp. 987–1007. DOI: 10.2307/1912773. URL: https://doi.org/10.2307/1912773.

Hamilton, James D. (1989). 'A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle'. In: *Econometrica* 57.2, pp. 357–384. DOI: 10.2307/1912559. URL: https://doi.org/10.2307/1912559.

— (1994). *Time Series Analysis*. Princeton University Press. URL: https://press.princeton.edu/books/hardcover/9780691042893/time-series-analysis.

Hastie, Trevor, Robert Tibshirani and Jerome Friedman (2009). *The Elements of Statistical Learning*. 2nd ed. Springer. ISBN: 978-0387848570.

Hautsch, Nikolaus (2012). *Econometrics of Financial High-Frequency Data*. Berlin, Heidelberg: Springer. ISBN: 978-3-642-21925-2. DOI: 10.1007/978-3-642-21925-2.

HistData.com (2026). *Free Forex Historical Data*. Accessed 2026-02-09. HistData.com. URL: https://www.histdata.com/ (visited on 09/02/2026).

Jacod, Jean et al. (2017). 'Between Data Cleaning and Inference: Pre-averaging and Robust Estimators of the Efficient Price'. In: *Journal of Econometrics* 196.1, pp. 1–22. DOI: 10.1016/j.jeconom.2016.08.014. URL: https://doi.org/10.1016/j.jeconom.2016.08.014.

Lee, Donny (2013). *Trading USDCHF Filtered by Gold Dynamics via HMM Coupling*. arXiv: 1308.0900 [stat.ML]. URL: https://arxiv.org/abs/1308.0900.

Lo, Andrew W. and A. Craig MacKinlay (1990). 'An Econometric Analysis of Nonsynchronous Trading'. In: *Journal of Econometrics* 45.1-2, pp. 181–211. DOI: 10.1016/0304-4076(90)90098-E. URL: https://doi.org/10.1016/0304-4076(90)90098-E.

Rabiner, Lawrence R. (1989). 'A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition'. In: *Proceedings of the IEEE* 77.2, pp. 257–286. DOI: 10.1109/5.18626. URL: https://doi.org/10.1109/5.18626.

Tsay, Ruey S. (2010). *Analysis of Financial Time Series*. 3rd ed. Wiley. URL: https://www.wiley.com/en-us/Analysis+of+Financial+Time+Series%2C+3rd+Edition-p-9780470414354.

Zabaljauregui, Diego and Luciano Campi (2020). 'Optimal Market Making under Partial Information with General Intensities'. In: *Applied Mathematical Finance*. arXiv: 1902.01157. URL: https://arxiv.org/abs/1902.01157.